

# torch.pow#

<https://pytorch.org/docs/stable/generated/torch.pow.html>

## Description:

Takes the power of each element in input with exponent and returns a tensor with the result. exponent can be either a single float number or a Tensor with the same number of elements as input. When exponent is a scalar value, the operation applied is:  $\text{input}^{\text{exponent}}$ . When exponent is a tensor, the operation applied is:  $\text{input}^{\text{exponent}}$ . When exponent is a tensor, the shapes of input and exponent must be broadcastable.

## Function Signature

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```
torch.pow(input, exponent, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

```
torch.pow(self, exponent, *, out=None) → Tensor
```

Parameters

## Parameters

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### Keyword

out (Tensor, optional) – the output tensor.

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## Code Examples

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```
>>> a = torch.randn(4)
>>> a
tensor([ 0.4331,  1.2475,  0.6834, -0.2791])
>>> torch.pow(a, 2)
tensor([ 0.1875,  1.5561,  0.4670,  0.0779])
>>> exp = torch.arange(1., 5.)
```

```
>>> a = torch.arange(1., 5.)
>>> a
tensor([ 1.,  2.,  3.,  4.])
>>> exp
tensor([ 1.,  2.,  3.,  4.])
>>> torch.pow(a, exp)
tensor([  1.,  4., 27., 256.])
```

```
>>> exp = torch.arange(1., 5.)
>>> base = 2
>>> torch.pow(base, exp)
tensor([ 2.,  4.,  8., 16.])
```

# Detailed Documentation

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## Documentation

Takes the power of each element in input with exponent and returns a tensor with the result.

exponent can be either a single float number or a Tensor with the same number of elements as input.

When exponent is a scalar value, the operation applied is:

When exponent is a tensor, the operation applied is:

When exponent is a tensor, the shapes of input and exponent must be broadcastable.

input (Tensor) – the input tensor.

exponent (float or tensor) – the exponent value

out (Tensor, optional) – the output tensor.

self is a scalar float value, and exponent is a tensor. The returned tensor out is of the same shape as exponent

The operation applied is:

self (float) – the scalar base value for the power operation

exponent (Tensor) – the exponent tensor

out (Tensor, optional) – the output tensor.

